

Sturmian Sequences

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1 Windows and splits

Take an infinite sequence of 0s and 1s:

$$\ell = 0100101001001010\dots$$

Slide a window m letters wide along it and write down every distinct block you see through it. Let $F(m)$ be how many distinct blocks there are.

Do it. Take $m = 3$ and walk the window along the sixteen letters above:

position	window	block	
0	010 0101001001010	010	new
1	0 100 101001001010	100	new
2	01 001 01001001010	001	new
3	010 010 1001001010	010	seen
4	0100 101 001001010	101	new
5	01001 010 01001010	010	seen
6	010010 100 1001010	100	seen
7	0100101 001 001010	001	seen
8	01001010 010 01010	010	seen
$\dots$ and nothing new appears after that $\dots$			

Four distinct blocks, so $F(3) = 4$. Note which of the eight possible three-letter blocks never appear at all: 110, 011, 111 and 000. This sequence is far from arbitrary.

Now widen the window by one letter and ask what becomes of each block. Do it: for each of the four, collect every letter seen following it anywhere in the sequence.

block	occurs at	followed by	becomes
010	0, 3, 5, 8, 11, 13	0 at 0, 1 at 3	0100 and 0101
100	1, 6, 9	1, every time	1001
001	2, 7, 10	0, every time	0010
101	4, 12	0, every time	1010

Three of the four blocks pass through unchanged. One does not. **010** sits at position 0 followed by a 0, and at position 3 followed by a 1, so it becomes two blocks. Call such a block *right-special*, and say it *splits*.

The blocks of length four are therefore

$$\mathbf{0010}, \quad \mathbf{0100}, \quad \mathbf{0101}, \quad \mathbf{1001}, \quad \mathbf{1010},$$

five of them, so $F(4) = 5$. We started with four and exactly one split — and the two blue ones are the two halves of it.

Every block either passes through or splits in two, so

Definition 1.

$$F(m+1) - F(m) = \#\{\text{blocks of length } m \text{ that split}\}.$$

Here is why the count is worth having. Suppose you are reading ℓ one letter at a time, trying to call the next letter, and you can remember only the last m . Then two occurrences of the same block are indistinguishable to you: you are in the same situation both times, and whatever rule you are using must say the same thing at both. If that block splits, the next letter genuinely differs between the two occurrences, so you are wrong at one of them. Not through bad judgement — the letters you can see do not contain the answer.

So counting splits counts the situations in which m letters of memory are not enough, and a sequence whose splitting stops at some finite m is one you can predict perfectly from a finite table.

2 Sturmian sequences

Definition 2. A sequence is *Sturmian* when exactly one block of each length splits:

$$F(m) = m + 1 \quad \text{for every } m \geq 1.$$

The sequence of §1 is Sturmian: it is the Fibonacci sequence, the fixed point of $0 \mapsto 01, 1 \mapsto 0$. We found $F(3) = 4$ and $F(4) = 5$, with **010** the block that splits. Here are two more, worked the same way.

Example 3 (the Pell sequence, $0 \mapsto 001, 1 \mapsto 0$).

$$\ell = 001001000100100010010010 \dots$$

Its four blocks of length three are **000**, **001**, **010**, **100**, so $F(3) = 4$. Note 11 never occurs.

block	occurs at	followed by	becomes
000	6, 13, 23	1, every time	0001
001	0, 3, 7, 10	0, every time	0010
010	1, 4, 8, 11	0, every time	0100
100	2, 5, 9, 12	1 at 2, 0 at 5	1001 and 1000

One split, so $F(4) = 5$: **0001**, **0010**, **0100**, **1000**, **1001**.

Example 4 (the rotation of slope $\alpha = 1/\sqrt{2}$).

$$\ell = 011011011101101110110110 \dots$$

Now the 1s are in the majority — the density is 0.707 — and 00 never occurs. The four blocks of length three are **011**, **101**, **110**, **111**.

block	occurs at	followed by	becomes
011	0, 3, 6, 10	0 at 0, 1 at 6	0110 and 0111
101	2, 5, 9, 12	1, every time	1011
110	1, 4, 8, 11	1, every time	1101
111	7, 14, 24	0, every time	1110

Again one split, so $F(4) = 5$: **0110**, **0111**, **1011**, **1101**, **1110**.

Three sequences with nothing in common to look at — one sparse, one balanced, one dense; two built by substitution and one by rotation — and in each the same thing happens. Four blocks of length three, exactly one of them ambiguous, five blocks of length four. Whatever the slope, there is precisely one situation at each width in which the letters you can see do not determine the next one.

Those three are not a coincidence, and there is a complete description of the sequences that behave this way.

Theorem 5 (Morse–Hedlund). *A sequence ℓ is Sturmian if and only if there are an irrational $\alpha \in (0, 1)$ and a $\rho \in [0, 1)$ with*

$$\ell_k = \lfloor (k+1)\alpha + \rho \rfloor - \lfloor k\alpha + \rho \rfloor \quad \text{for all } k \geq 0.$$

The slope α is the density of 1s in ℓ and is determined by ℓ ; the intercept ρ only says where in the sequence you started reading. We do not prove this.

So there is one Sturmian sequence for each irrational slope, up to where you begin. The three worked above are

sequence	slope α			
Fibonacci, $0 \mapsto 01, 1 \mapsto 0$	$1/\varphi^2 = (3 - \sqrt{5})/2$	≈ 0.381966	no 11	
Pell, $0 \mapsto 001, 1 \mapsto 0$	$1 - 1/\sqrt{2}$	≈ 0.292893	no 11	
the rotation of §2	$1/\sqrt{2}$	≈ 0.707107	no 00	

The last column is not an accident either: a Sturmian sequence contains 11 exactly when $\alpha > \frac{1}{2}$, and 00 exactly when $\alpha < \frac{1}{2}$. Two of these slopes lie below a half and one above.

3 Why Sturmian sequences make interesting monsters

Here is a game built from one of these sequences. It is the monster’s schedule: at beat k the monster strikes if $\ell_k = 1$, and stands open if $\ell_k = 0$.

You never see the current letter before you commit. You see only the letters already past. Each beat you choose one of two things:

block nothing happens, whatever the monster does
attack on an open beat, the monster loses 2
 into a strike, you lose 5

You start on 20 points and the monster on 60. The fight lasts 120 beats.

Three things follow. You need thirty clean reads to win. You can afford three mistakes. And blocking forever is a loss, not a draw — the clock runs out with the monster untouched.

A player who remembers the last m letters is exactly the thing §1 describes: it cannot tell two occurrences of the same block apart, so it plays them identically, and the blocks that split are precisely the situations where that costs it. Give such a player a table learned from previous encounters and set it against monsters of each kind, all with the same density of strikes so that only the splitting differs:

monster	$m = 2$	$m = 4$	$m = 6$	$m = 8$	$m = 12$
periodic, period 7	0%	14%	100%	100%	100%
Sturmian	9%	47%	58%	72%	86%
random	0%	0%	0%	0%	0%

Win rates over 300 fights each. Three shapes: a step, a climb, a flat line.

Periodic: nothing left. Its splitting stops at $m = 6$, and the table says the same thing. Below six letters the player has almost nothing; at six it has a complete theory of the monster and never loses again. There is one threshold, and on the far side of it the game is over permanently — not won, but finished, because no further observation can teach anything. A monster you beat this way you have beaten for good, and the second encounter is the first one replayed.

Random: nothing to get. Every block splits at every length, so no table of any depth is ever a theory of anything. Memory 12 is worth exactly what memory 2 is worth, which is nothing: the player is guessing, guessing costs 5 and pays 2, and it loses every time. Study is not merely insufficient here, it is pointless, and a player who works that out correctly concludes there is nothing to work out.

Sturmian: always something, never everything. Exactly one block of each length splits. So a player holding m letters has exactly one situation it cannot call — and by the nesting of §2, the ambiguous block at depth $m + 1$ sits inside the one at depth m . Each extra letter of memory therefore cuts the outstanding ambiguity into a rarer case without ever removing it. That is precisely a curve which rises at every depth and reaches 100% at none, and it is what the middle row measures.